intensity, sound pressure, or an infinite number of other parameters. Since we don't know what it represents in this particular case, we will give it the generic label: **amplitude**. This parameter is also called several other names: the **y-axis**, the **dependent variable**, the **range**, and the **ordinate**.

The **horizontal axis** represents the other parameter of the signal, going by such names as: the **x-axis**, the **independent variable**, the **domain**, and the **abscissa**. *Time* is the most common parameter to appear on the horizontal axis of acquired signals; however, other parameters are used in specific applications. For example, a geophysicist might acquire measurements of rock density at equally spaced *distances* along the surface of the earth. To keep things general, we will simply label the horizontal axis: **sample number**. If this were a continuous signal, another label would have to be used, such as: *time*, *distance*, *x*, etc.

The two parameters that form a signal are generally not interchangeable. The parameter on the y-axis (the dependent variable) is said to be a **function** of the parameter on the x-axis (the independent variable). In other words, the independent variable describes *how* or *when* each sample is taken, while the dependent variable is the actual measurement. Given a specific value on the x-axis, we can always find the corresponding value on the y-axis, but usually not the other way around.

Pay particular attention to the word: *domain*, a very widely used term in DSP. For instance, a signal that uses time as the independent variable (i.e., the parameter on the horizontal axis), is said to be in the **time domain**. Another common signal in DSP uses frequency as the independent variable, resulting in the term, **frequency domain**. Likewise, signals that use distance as the independent parameter are said to be in the **spatial domain** (distance is a measure of space). The type of parameter on the horizontal axis *is* the domain of the signal; it's that simple. What if the x-axis is labeled with something very generic, such as *sample number*? Authors commonly refer to these signals as being in the *time* domain. This is because sampling at equal intervals of time is the most common way of obtaining signals, and they don't have anything more specific to call it.

Although the signals in Fig. 2-1 are discrete, they are displayed in this figure as continuous lines. This is because there are too many samples to be distinguishable if they were displayed as individual markers. In graphs that portray shorter signals, say less than 100 samples, the individual markers are usually shown. Continuous lines may or may not be drawn to connect the markers, depending on how the author wants you to view the data. For instance, a continuous line could imply what is happening *between* samples, or simply be an aid to help the reader's eye follow a trend in noisy data. The point is, examine the labeling of the horizontal axis to find if you are working with a discrete or continuous signal. Don't rely on an illustrator's ability to draw dots.

The variable, N, is widely used in DSP to represent the total number of samples in a signal. For example, $N = 512$ for the signals in Fig. 2-1. To